Identify the two parts

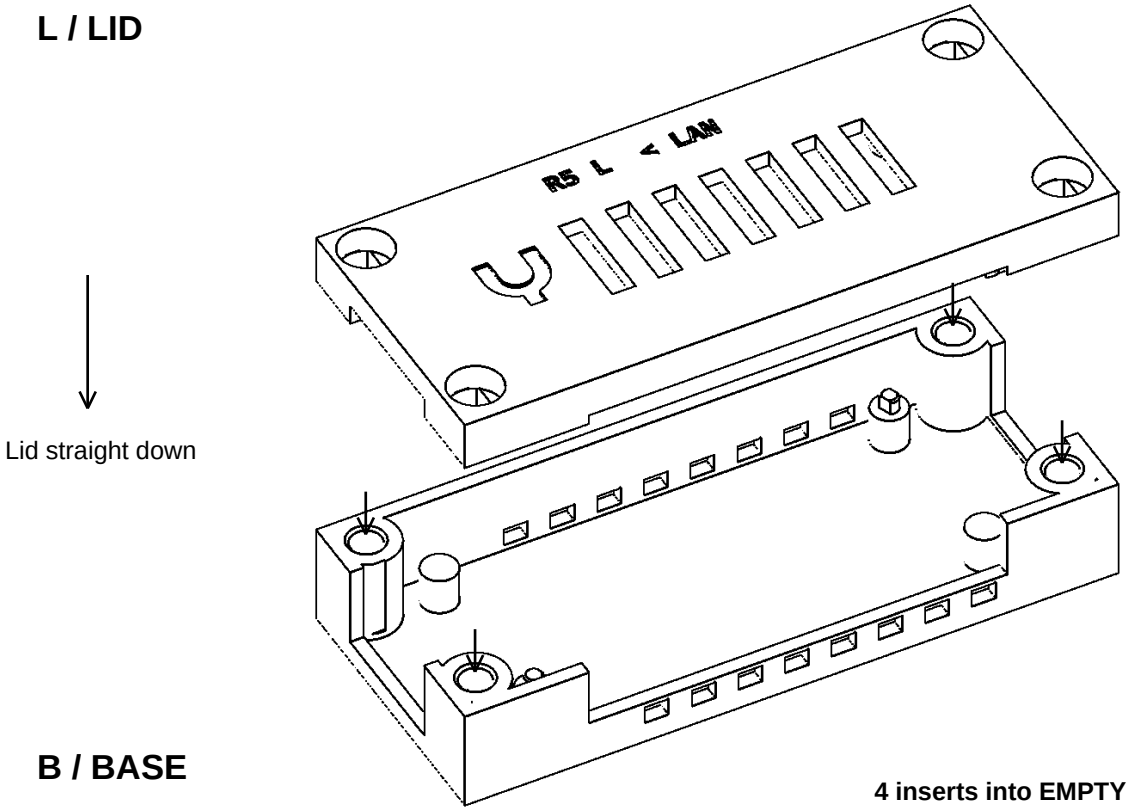
R5 is a 84 x 35.9 x 20 rectangular, ventilated case for a bare Radxa ZERO 3E. The lid lowers vertically onto the base; all four screws enter from above.

Inventory

1 x R5 B base | 1 x R5 L lid | 1 x bare ZERO 3E board
4 x M3 x 8 socket-head screws | 4 x RX-M3x5.7 heat-set inserts

Screw envelope: 8 under head; head no larger than diameter 6 x height 3. Insert: diameter 4.6 x length 5.7; nominal printed pilot diameter 4.0.

L / LID



4 inserts into EMPTY base

Both LAN arrows point to Ethernet (left in these views).

1 Match the R5 parts and orientation

Identify R5 B base and R5 L lid. Keep both LAN arrows at the Ethernet end. Use only the matched R5 pair and the bare board.

2 Check the hardware before fitting it

Check four M3 x 8 socket-head screws and four RX-M3x5.7 inserts against the listed nominal envelopes. Confirm the insert stock label and measure stock before heating. Check driver and heater access with the actual tools.
Stop if the stock identity, dimensions, pilot, or tool access differs; do not substitute hardware by label alone.

Tools: caliper/depth gauge, correct socket driver, controlled insert heater and tip, nonflammable support; actual Ethernet and both USB-C cables. Use a host and temperature monitoring for later powered observations.

Fit empty, then install inserts

3 Test the empty nested joint

With no board, inserts, or screws fitted, orient the two LAN ends together and lower the lid straight down. It must nest and seat smoothly by hand. Lift vertically to separate.

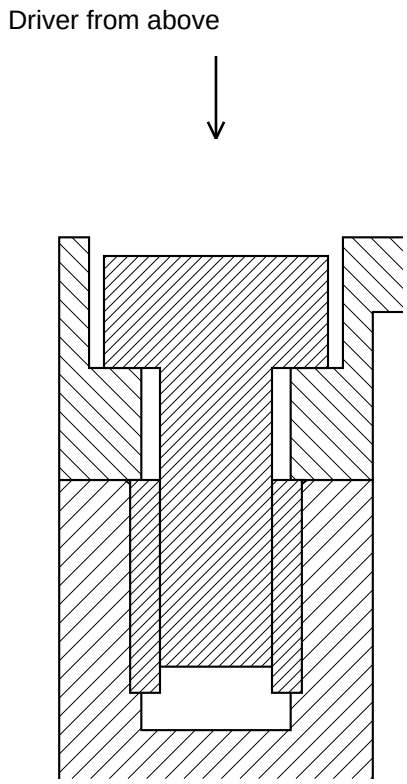
Stop for a blocked, skewed, or forced fit. Do not pull the halves together with screws.

4 Heat-set four inserts in the EMPTY base

Support the EMPTY base flat on a nonflammable surface. Heat-set one checked insert vertically into each of the four pilots, flush with its boss top. Use the insert maker's method and a controlled heater. Let the base cool fully. Check every insert is flush with its boss top, never proud, before installing the board.

Stop for a tilted insert, lifted boss, crack, blocked thread, or distortion. Do not heat-set with the board installed.

A-A / front-left screw axis / Y = 1.6 / assembled section



All four joints use the same stack

20.0	exterior lid surface
19.5	nominal head top (0.5 recessed)
16.5	head bearing land
13.5	insert top / base boss stop
8.5	screw tip (8 below its head)
7.8	insert bottom
6.8	blind pilot bottom

Load path: screw head -> lid bearing land -> base boss -> insert thread. The PCB is outside this clamp stack.

Nominal axial overlap 5.0; tip reserve 0.7 to insert bottom and 1.7 to pilot bottom. Confirm actual screw and insert lengths before fitting.

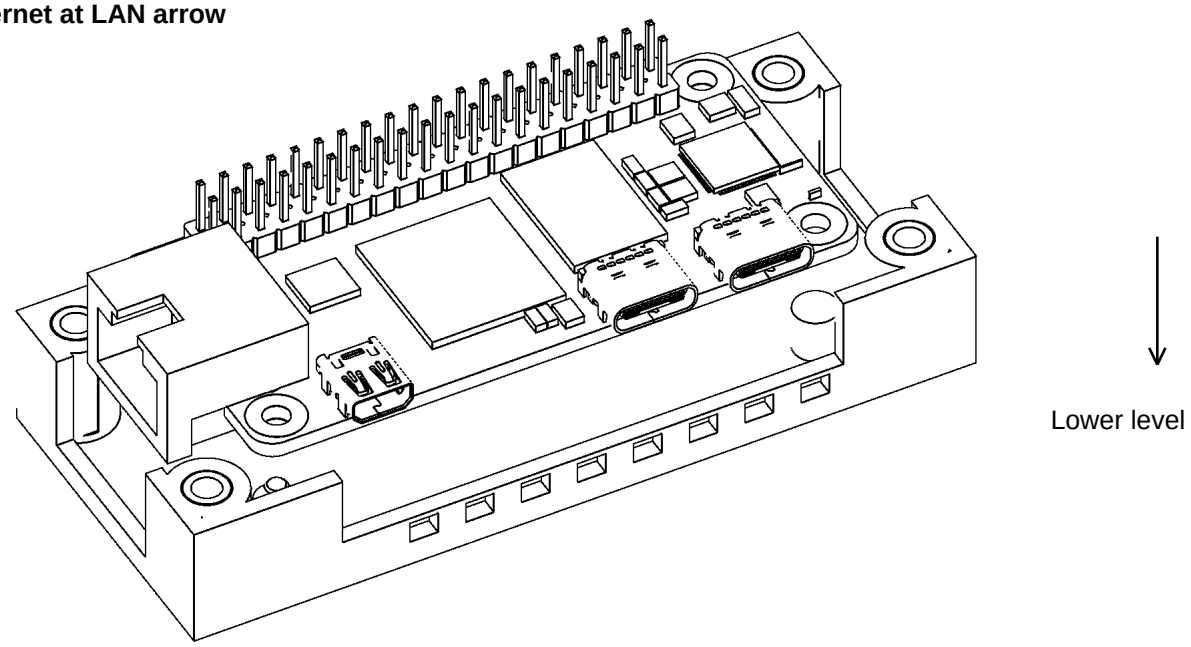
Plastic: opposite hatches. Metal: close hatching.

Section comes from the actual base, lid, insert and screw meshes. Nominal hardware envelopes are shown; threads are not represented. Driver access is vertical through the recessed head well. Check the actual driver before closing.

The lid skirt nests with 0.4 clearance per side and a chamfered entry (section B-B, page 4). Clear only loose print debris; a changed fit or damaged boss returns to design.

Lower, close and reopen

Bare board / Ethernet at LAN arrow



5 Lower the unpowered board

Disconnect power and all cables. If using microSD, install the card before seating the board. Put the bare board's Ethernet socket at the LAN end; the two USB-C sockets face the front openings. Lower the board vertically and level onto its pads and two locators. Check that all components and connectors clear.

Stop if the board rocks, bows, catches, or rests on a component instead of the support pads.

6 Lower the lid straight down

Match the LAN ends. Lower the lid vertically over the base. The nested lip locates the lid; it does not slide or hook. Check an even, seated seam before adding screws.

Stop if the lid binds or presses on the board. Screws must not fix a poor fit.

7 Start all four screws, then snug

Start all four M3 x 8 screws by hand from above. Lightly snug opposite corners in turn with the socket driver. Heads must remain below the lid surface; do not overtighten or bow the lid.

Stop for early bottoming, cross-threading, an insert turning, or a head that stands proud.

8 To reopen: screws out, then lift

Disconnect power and cables. Remove all four screws completely, lift the lid straight up, then lift the board vertically off the locators. Keep the base supported flat.

Do not pry against the PCB or connectors, and do not slide the lid sideways.

Check the first full case

Record observations on the real parts. A digital fit check does not establish a physical fit. The first article is an unloaded fit and service trial; thermal qualification remains open.

A Check parts, board and actual cables

Inspect the empty fit, cooled inserts, board support, seam, screw recesses, and each actual Ethernet/USB-C plug and cable collar. For the recessed microSD card, use the open/lift-board service sequence if external access is awkward. Hold the closed, unpowered case gently in its ordinary use orientations. Small board play is designed in; record actual play/rattle and any loosening.

Stop for connector/collar contact, bowing, rocking off supports or locators, unstable support, cracks, or strained cables. Actual plugs, collars and microSD fit have not been measured by the model.

B Open and close ten times

With power and cables disconnected, repeat the complete open/close sequence ten times. Each closure must seat freely before tightening. Recheck the seam, locators, inserts, screw threads, and cable access.

Stop for a new bind, crack, loose insert, damaged thread, or changed board fit. Record the cycle and return the failed interface to design.

C Observe heat in the intended setup

Stand the base on a hard, flat desk with the lid up. Keep the low side inlets and top outlets open; do not cover the case or place it on soft fabric. Vent paths are modeled, but cooling performance is unmeasured.

After fit checks pass, monitor the actual intended workload. Record ambient, CPU and case temperatures, elapsed time, throttling and resets. Stop for abnormal heat, distortion, odor, unstable operation, or a component maker's operating limit. No safe power or temperature rating is established by this guide.

Record before deciding to use the case

Date / material / printer: _____

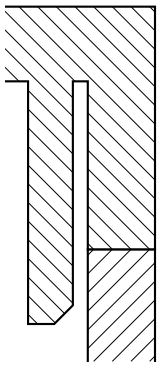
Insert stock / measured OD x length: _____

Fit / cables / cycles / issues: _____

Workload / ambient / temperatures / time: _____

Outcome / failed interface / next action: _____

B-B / rear side joint at X = 22 / Y horizontal, Z up



L / lid and its inner skirt

B / base wall

Nominal per-side gap 0.4; nesting depth 2; entry chamfer 0.5. Opposite hatches distinguish the printed parts. No hook or sideways assembly motion.

Fit and reopening must be observed on the printed pair before screws are used. Keep the same Ethernet end throughout assembly.